47202 I.F.E. Problems, Lecture 7 2026
Solutions

Short questions

1. What is the role of the electrolyte in an electrochemical cell?

To conduct ions between the electrodes.

To separate the electrodes electronically.

To separate the reactants (and the products).

2. Which properties must an electrolyte have?

High ionic conductivity, high electronic resistivity, low gas permeability in case gases are involved, thermal and chemical stability. Moreover, it should be a stable physical barrier to prevent cell short circuit. Finally, we like it to be inexpensive.

3. How are the anode and the cathode defined?

The anode is the electrode at which an oxidation takes place (a species increases the oxidation state at the anode).

The cathode is the electrode at which a reduction takes place (a species decreases its oxidation state at the cathode).

4. What is the basic difference between thermodynamics and kinetics?

Thermodynamics determines if a process is possible at all and quantifies the energy liberated or consumed. It represents the ideal case and does not take losses into account.

Kinetics determines how easily and how fast a possible process is carried out. It quantifies reaction rate and losses. A catalyst can improve kinetics but never change thermodynamics of a process.

In brief: thermodynamics tells us what is possible; kinetics tells us how fast.

Also in brief: thermodynamics determines the maximum energy; Kinetics determines the actual power.

5. Balance the following half-cell reactions

$PbSO_4 \rightleftarrows Pb + SO_4^{2-}$ (one of the electrode reactions in a lead-acid battery) (don't look in the text😉)

All the atoms fit already. Only the charge needs to be balanced by electrons:

$PbSO_4 + 2e^- \rightleftarrows Pb + SO_4^{2-}$

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$Ni(OH)_2 \rightleftarrows NiOOH + H^+$ (the nickel electrode in some batteries)

Atoms fit. Balance charge with electrons:

$Ni(OH)_2 \rightleftarrows NiOOH + e^- + H^+$

$O_2 + H_2O \rightleftarrows OH^-$ (the oxygen process in some fuel cells). Note that with OH^-, the system is alkaline and there should be no protons (H^+) in the equation (Protons, if any, would immediately react with OH^- and form water).

The non-chemical approach: 3O on the left side require $3OH^-$ on the right.

$O_2 + H_2O \rightleftarrows 3OH^-$

Then H doesn't fit. Add one more H_2O on the left and then you have 4O and 4H on the left, and this is balanced by $4OH^-$ on the right.

$O_2 + 2H_2O \rightleftarrows 4OH^-$

Then balance the charge with electrons

$O_2 + 2H_2O + 4e^- \rightleftarrows 4OH^-$

The chemical approach: O_2 is normally considered to be reduced to O^{2-} also if it appears in H_2O or OH^-. Reduction of O_2 thus requires $4e^-$

$O_2 + H_2O + 4e^- \rightleftarrows OH^-$

The charge is balanced by the same amount of OH^-

$O_2 + H_2O + 4e^- \rightleftarrows 4OH^-$

The atoms are balanced by one more H_2O

$O_2 + 2H_2O + 4e^- \rightleftarrows 4OH^-$

$MnO_4^- + H^+ \rightleftarrows MnO_2 + H_2O$ (the cathode process in some batteries). H^+ means an acidic environment and therefore OH^- should not be there.

Start by trying one Mn on each side as already written. Then the two leftover O atoms need to form $2H_2O$ and this takes four protons.

$MnO_4^- + 4H^+ \rightleftarrows MnO_2 + 2H_2O$

Balance with electrons

$MnO_4^- + 4H^+ + 3e^- \rightleftarrows MnO_2 + 2H_2O$

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Note: a chemist would see that Mn change form oxidation state +7 to +4 and this requires 3 electrons. However, you don't need to know that to balance the equation.

6. Which combination of reactions from the table with *Standard potentials* (E^o) in the text will give the highest cell voltage? How high is it?

$Li^+ + e^- \rightleftarrows Li$ -3,01 V

$F_2 + 2e^- \rightleftarrows 2F^-$ 2,87 V

Cell voltage: 2,87 V – (-3,01 V) = 5.88 V

7. Can we make such a battery? Would it be a superior battery?

In principle, yes, but there may be several problems. 1) We need an electrolyte that is stable in such a high potential difference. 2) F_2 is a gas. The electrolyte must be impermeable to F_2 gas. If not, the battery will self-discharge or even explode. 3) F_2 as a gas has to be stored and it takes up much space even at a high pressure. A high-pressure tank is heavy. Besides, corrosion problems can be expected. It is probably not a good idea after all.

Calculations:

8. What would the standard cell potential be if a hydrogen/oxygen fuel cell were operated on hydrogen and chlorine gas instead?

Hydrogen/oxygen:

Cathode: $O_2 + 4H^+ + 4e^- \rightleftarrows 4H_2O$ 1.23 V

Anode: $2H^+ + 2e^- \rightleftarrows H_2$ (reversed) 0.0000 V

Total cell: 1.23 V - 0.0000 V = 1.23 V

Hydrogen/chlorine:

Cathode: $Cl_2 + 2e^- \rightleftarrows 2Cl^-$ 1.36 V

Anode: $2H^+ + 2e^- \rightleftarrows H_2$ (reversed) 0.0000 V

Total cell: 1.36 V - 0.0000 V = 1.36 V

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9. On fast charging. Consider a battery with the fixed internal ohmic resistance, *R,* of the electrolyte and with catalysts in the electrodes so efficient that the kinetic overpotential is negligible at any practical current. Can such a battery be fast charged without a significant loss of efficiency? Show why/why not?

The passage of the current through the electrolyte causes a voltage drop according to ohms law. $E = R \cdot I$.

Assuming 100% Faradaic efficiency and perfect kinetics, a higher current always gives a higher voltage drop. Thus, a higher charging voltage is necessary and since voltage times charge is energy, more energy is used to store the same electrical charge when charging the battery.

10. The common alkaline battery is made from manganese and zinc and the electrode discharge reactions are:

Anode: $Zn(s) + 2OH^-(aq) \rightleftarrows ZnO(s) + H_2O(l) + 2e^-$

Cathode: $2MnO_2(s) + H_2O(l) + 2e^- \rightleftarrows Mn_2O_3(s) + 2OH^-(aq)$

a) What is the total cell reaction?

Just add the left side, add the right sides and remove species that appear on both sides

Anode: $Zn(s) + \cancel{2OH^-(aq)} \rightleftarrows ZnO(s) + \cancel{H_2O(l)} + \cancel{2e^-}$

Cathode: $2MnO_2(s) + \cancel{H_2O(l)} + \cancel{2e^-} \rightleftarrows Mn_2O_3(s) + \cancel{2OH^-(aq)}$

$$Zn(s) + 2MnO_2(s) \rightleftarrows ZnO(s) + Mn_2O_3(s)$$

b) Calculate the standard cell potential. The standard free energy of formation can be found in the following table.

25 ºC, 1 bar abs.	Zn(s)	ZnO(s)	$MnO_2(s)$	$Mn_2O_3(s)$	$H_2O(l)$
Δg_f^0 (kJ mol^{-1})	0.0	-320.5	-465.1	-881.1	-137.1

$E_{cell} = \frac{-\Delta g_r}{nF}$, n = 2 (see half-cell reactions or consider: Zn → ZnO, O= -2)

$\Delta g_r = \Delta g_f(ZnO) + \Delta g_f(Mn_2O_3) - \Delta g_f(Zn) - 2\Delta g_f(MnO_2)$

Δg_r = -320500 + (-881100) - 0.0 – 2(-465100) = -271,400 J mol^{-1}

$$E_{cell} = \frac{-\Delta g_r}{nF} = \frac{271400\ \text{J mol}^{-1}}{2 \cdot 96485\ \text{C mol}^{-1}} = 1.41\ \text{V}$$

Comment: Since all species in the total equation are pure solids, their activities are 1 and the calculated potential should be what is seen from normal alkaline batteries (no need for Nernst). However, alkaline batteries normally have a cell voltage around 1.5 V, so there is apparently more to this story. MnO_2 can attain several different crystal structures, and if the material in the battery has a less stable structure than the most stable one in the thermodynamic table, then this may explain the last 0.1 V.

11. A high temperature fuel cell is operated on hydrogen and oxygen at 800 ºC. At this temperature, the free energy of formation of hydrogen, oxygen and water are:

800 ºC, 1 bar abs.	H_2	O_2	H_2O
Δg_f (kJ mol^{-1})	0.0 kJ mol^{-1}	0.0 kJ mol^{-1}	-188.5 kJ mol^{-1}

a) What is the standard cell potential at this temperature? The reaction is $H_2 + ½O_2 \rightleftarrows H_2O$

$\Delta g_r = \Delta g_f(H_2O) - \Delta g_f(H_2) - ½\Delta g_f(O_2)$

Δg_r = -188.5 kJ mol^{-1} – 0 - ½·0 = -188.5 kJ mol^{-1}

$$E_{cell} = \frac{-\Delta g_r}{nF} = \frac{188500\ \text{J mol}^{-1}}{2 \cdot 96485\ \text{C mol}^{-1}} = 0.98\ \text{V}$$

b) What is the reversible cell potential, if the cell is operated on hydrogen and air, each supplied at the pressure 2 bar absolute. Assume that the partial pressure of the produced water is 0.5 bar absolute and that the content of oxygen in air is 20%.

Use Nernst for a galvanic cell:

$$E_{\text{cell}} = E^{\text{o}}_{\text{cell}} - \frac{RT}{nF}\ln\frac{a_{\text{H2O}}}{a_{\text{H}_2}a_{\text{O}_2}^{½}} \approx E^{\text{o}}_{\text{cell}} - \frac{RT}{nF}\ln\frac{p_{\text{H2O}}}{p_{\text{H}_2}p_{\text{O}_2}^{½}}$$

(Remember, products in the numerator and reactants in the denominator)

The fraction of oxygen in air is 0.2:

$$E_{\text{cell}} \approx 0.98\ V - \frac{8.314\cdot(800+273.15)}{2\cdot 96485}\ln\frac{0.5}{2\cdot(2\cdot 0.2)^{½}} = \underline{1.023\ \text{V}}$$

12. A single cell in an industrial electrolyzer is operated for an hour with the current 2000 A. During that period, it produces 72 g hydrogen. The process is

$$H_2O \rightarrow H_2 + ½O_2.$$

Calculate the faradic efficiency of the process. Remember that each mole H_2 requires 2 mole electrons.

Faraday's law says $\dot{n}_{\text{con}} = \frac{I}{nF}$

I = 2000 A = 2000 C s^{-1}, n = 2, F = 96485 C mol^{-1}

$\dot{n}_{\text{con}} = \frac{I}{nF} = \frac{2000\ \text{C s}^{-1}}{2\cdot\ 96485\ \text{C mol}^{-1}}$ = 0.01036 mol s^{-1} = 0.01036 mol s^{-1} · 3600 s h^{-1}

= <u>37.31 mol h^{-1}</u>

37.31 mol h^{-1} corresponds to 37.31 mol h^{-1} · 2 g mol^{-1} = <u>74.62 g h^{-1}</u>

With an actual production rate of 72 g h^{-1} the faradaic efficiency becomes

$\eta = \frac{72}{74.62}$ = 0.96 or <u>96%</u>

Note: This production rate doesn't look much, but real industrial electrolyzer are stacks of many cells in series. The same current passes all cells. With 300 cells the production is 21.6 kg hydrogen per hour – enough to fill 5 small hydrogen vehicles.